

MID REVISION

ADVANCED NETWORK

Advanced Networking Mid-Revision Booklet

INSTRUCTOR

Eng Ibrahim Gamal

Contact: **01146987580**

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Part 1 – Core Concepts Review

Original page title: Advanced Network Midterm Revision 2025 · 14 multiple-choice questions · Networking basics, delays, switching and the transport layer

Q **Which is not an application layer protocol?**

A HTTP **B** SMTP **C** TCP **D** FTP

Q **For a 100Mbps Ethernet link, if the length of the packet is 32bits, the transmission delay is ___ (in microseconds)**

A 320 **B** 3.2 **C** 32 **D** 0.32

Q **In which one of the following switching, the bandwidth is variable?**

A Packet switching
B Both Circuit switching and Packet switching
C None of them
D Circuit switching

Q **In a network, If P is the only packet being transmitted and there was no earlier transmission, which of the following delays could be zero?**

A Queuing delay **B** Transmission delay
C Propagation delay **D** Processing delay

Q **Transmission delay does not depend on _____**

A Bandwidth of medium **B** Packet length
C Transmission rate **D** Distance between the routers

Q **What does TCP use to ensure reliable delivery of data?**

A Sequence numbers **B** All of them
C Checksums **D** Acknowledgments

Q **What is the primary purpose of the transport layer?**

A To route packets between devices **B** To manage physical connections
C To provide reliable data transfer **D** To handle application data formats

Q **In the context of the transport layer, what is a socket?**

A A type of network protocol
B A hardware interface
C A method for packet routing
D A software endpoint for communication

Q **Which transport layer protocol is connectionless?**

A FTP **B** UDP **C** TCP **D** SCTP

Q **Given L = number of bits in the packet, a = average rate, and R = transmission rate. The Traffic intensity in the network is provided by**

A $\frac{Ra}{L}$ **B** $\frac{R}{La}$ **C** $\frac{LR}{a}$ **D** $\frac{La}{R}$

Q **Which of the following delay is faced by the packet in travelling from one end system to another?**

A Propagation delay **B** Queuing delay
C Transmission delay **D** All of the mentioned

Q **For a 10Mbps Ethernet link, if the length of the packet is 32bits, the transmission delay is _____ (in microseconds)**

A 3.2 **B** 32 **C** 0.32 **D** 320

Q **The time required to examine the packet's header and determine where to direct the packet is part of _____**

A Processing delay
B Queuing delay
C Transmission delay
D Propagation delay

Q **The total nodal delay is given by _____**

A $d_{nodal} = d_{proc} - d_{queue} + d_{trans} + d_{prop}$
B $d_{nodal} = d_{proc} + d_{trans} - d_{queue}$
C $d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$
D $d_{nodal} = d_{proc} + d_{queue} - d_{trans} - d_{prop}$

Part 2 – Foundations, Models & Security

48 multiple-choice questions · Original numbering retained (the source file contains no Questions 41 and 49)

Network Fundamentals & Devices (Q1–Q10)

1 **What defines a computer network?**

A A device that shows output on a monitor
B A group of connected computers and devices that share resources and communicate
C Software for creating documents and presentations
D The outer casing that protects internal computer parts.

2 **Which device forwards data packets between networks by using routing information in the packet?**

A Firewall B Bridge C Hub D Router

3 **Which of the following is NOT a network edge device?**

A Switch B PC C Smartphone D Server

4 **What does a "router" primarily do in a computer network?**

A Routes data packets between different networks
B Filters network traffic for security
C Connects computers within a local network
D Converts analog signals to digital signals

5 **Protocols are a set of rules to govern _____**

A Communication B Standard
C Metropolitan communication D Bandwidth

6 **Which of the following is an example of Bluetooth?**

A wide area network B virtual private network
C local area network D personal area network

7 **When a collection of various computers appears as a single coherent system to its clients, what is this called?**

A mail system B networking system
C computer network D distributed system

8 **What is the term for an endpoint of an inter-process communication flow across a computer network?**

A port B machine C socket D pipe

9 **Which of these is not a network edge device?**

A Switch B PC C Smartphones D Servers

10 **Which of the following allows you to connect and login to a remote computer?**

A SMTP B HTTP C FTP D Telnet

Transmission Media (Q11–Q14)

11 **Which of these is not a guided media?**

A Fiber optical cable
B Coaxial cable
C Wireless LAN
D Copper wire

12 **Fiber optics possess the following properties _____**

A Immune electromagnetic interference
B Very less signal attenuation
C Very hard to tap
D All of the mentioned

13 **Coaxial cable consists of _____ concentric copper conductors.**

A 1 B 2 C 3 D 4

14 **Radio channels are attractive medium because _____**

A Can penetrate walls
B Connectivity can be given to a mobile user
C Can carry signals for long-distance
D All of the mentioned

Switching, Delay & Throughput (Q15–Q23)

15 **What are the Methods to move data through a network of links and switches?**

A Packet switching and Line switching
B Circuit switching and Line switching
C Line switching and bit switching
D Packet switching and Circuit switching

16 **As the resources are reserved between two communicating end systems in circuit switching, _____ is achieved.**

A authentication
B guaranteed constant rate
C reliability
D store and forward

17 **In which one of the following switching, the bandwidth is fixed?**

A Circuit switching
B Packet switching
C Both a and b
D None of the above

18 **The internet is an example for _____ circuit network?**

A Datagram subnet
B Virtual circuit subnet
C Both a and b
D None of the above

19 **The robustness to failures is more in _____ type of circuit?**

A Datagram subnet
B Virtual circuit subnet
C Both a and b
D None of the above

20 **Which one of the following circuits is connectionless?**

A Datagram subnet
B Virtual circuit subnet
C Both a and b
D None of the above

21 **For a 10Mbps Ethernet link, if the length of the packet is 32bits, the transmission delay is _____ (in microseconds)**

A 3.2 B 32 C 0.32 D 320

22 **Transmission delay does not depend on _____**

A Packet length
B Distance between the routers
C Transmission rate
D Bandwidth of medium

23 **In a network, If P is the only packet being transmitted and there was no earlier transmission, which of the following delays could be zero?**

A Propagation delay
B Queuing delay
C Transmission delay
D Processing delay

Layered Architecture: OSI & TCP/IP (Q24–Q36)

24 **How many layers are present in the Internet protocol stack (TCP/IP model)?**

A 5 B 7 C 6 D 10

25 **Which of the following layers is an addition to OSI model when compared with TCP IP model?**

A Application layer
B Presentation layer
C Session layer
D Session and Presentation layer

26 **The functionalities of the presentation layer include _____**

A Data compression
B Data encryption
C Data description
D All of the mentioned

27 **Delimiting and synchronization of data exchange is provided by _____**

A Application layer
B Session layer
C Transport layer
D Link layer

28 **In the OSI model, as a data packet moves from the lower to the upper layers, headers are _____**

A Added B Removed C Rearranged D Randomized

29 **Which of the following statements can be associated with OSI model?**

A A structured way to discuss and easier update system components
B One layer may duplicate lower layer functionality
C Functionality at one layer no way requires information from another layer
D It is an application specific network model

30 **Which layer is responsible for process to process delivery in a general network model?**

A network layer
B transport layer
C session layer
D data link layer

31 **Which address is used on the internet for employing the TCP/IP protocols?**

A physical address and logical address
B port address
C specific address
D all of the mentioned

32 **Which address is used on the internet for employing the TCP/IP protocols?**

A physical address and logical address
B port address
C specific address
D all of the mentioned

33 **Which address is used to identify a process on a host by the transport layer?**

A physical address
B logical address
C port address
D specific address

34 **Transmission data rate is decided by _____**

A network layer
B physical layer
C data link layer
D transport layer

35 **Encryption and Compression are functions of which OSI layer?**

A Presentation Layer
B Application Layer
C Session Layer
D Data Link Layer

36 **File, Transfer, Access and Management (FTAM) is a function of which layer?**

A Presentation Layer
B Application Layer
C Session Layer
D Data Link Layer

Network Security (Q37–Q40)

37 **The attacker using a network of compromised devices is known as _____**

A Internet
B Botnet
C Telnet
D D-net

38 **Which of the following is a form of DoS attack?**

A Vulnerability attack
B Bandwidth flooding
C Connection flooding
D All of the mentioned

39 **The DoS attack, in which the attacker establishes a large number of half-open or fully open TCP connections at the target host is _____**

A Vulnerability attack
B Bandwidth flooding
C Connection flooding
D UDP flooding

40 **Packet sniffers involve _____**

A Active receiver
B Passive receiver
C Legal receiver
D Partially-active receiver

History, Protocols & Performance (Q42–Q50)

42 **ARPANET stands for?**

A Advanced Research Project Agency Network.
B Advanced Research Programmed Auto Network.
C Advanced Research Project Automatic Network.
D Advanced Research Project Authorized Network.

43 **A technique that marked the beginning of computer communications.**

A Batch Environment
B Message passing
C User Environment
D Time Sharing

44 **TCP stands for?**

A Transmission control program
B Transmission control protocol
C Transfer control program
D Transfer control protocol

45 **Which is not a application layer protocol?**

A HTTP
B SMTP
C FTP
D TCP

46 **Given L = number of bits in the packet, a = average rate and R = transmission rate. The Traffic intensity in the network is given by _____**

A La/R
B LR/a
C R/La
D Ra/L

47 **In the transfer of file between server and client, if the transmission rates along the path is 10Mbps, 20Mbps, 30Mbps, 40Mbps. The throughput is usually _____**

A 20Mbps
B 10Mbps
C 40Mbps
D 50Mbps

48 **The total nodal delay is given by _____**

A $d_{nodal} = d_{proc} - d_{queue} + d_{trans} + d_{prop}$
B $d_{nodal} = d_{proc} + d_{trans} - d_{queue}$
C $d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$
D $d_{nodal} = d_{proc} + d_{queue} - d_{trans} - d_{prop}$

50 **Propagation delay depends on _____**

A Packet length
B Transmission rate
C Distance between the routers
D Speed of the CPU

Part 3 – Transport Layer

40 multiple-choice questions · TCP, UDP, SCTP, sockets, flow & congestion control

1 **what is the primary purpose of the transport layer?**

A To route packets between devices
B To provide reliable data transfer
C To manage physical connections
D To handle application data formats

2 **Which of the following protocols is used for reliable transport?**

A UDP B IP C TCP D ICMP

3 **What does TCP use to ensure reliable delivery of data?**

A Checksums
B Sequence numbers
C Acknowledgments
D All of the above

4 **In the context of the transport layer, what is a socket?**

A A hardware interface
B A software endpoint for communication
C A type of network protocol
D A method for packet routing

5 **Which transport layer protocol is connectionless?**

A TCP B SCTP C UDP D FTP

6 **What is the main advantage of using UDP over TCP?**

A Reliability
B Congestion control
C Lower latency
D Error recovery

7 **What term is used to describe the process of breaking down data into segments for transmission?**

A Fragmentation B Segmentation C Encapsulation D Multiplexing

8 **What is the purpose of flow control in TCP?**

A To manage the speed of data transmission
B To establish a connection
C To ensure data integrity
D To maintain session state

9 **In TCP, what is the function of the three-way handshake?**

A To synchronize data transmission
B To establish a reliable connection
C To detect packet loss
D To close a connection

10 **What is the maximum size of a TCP segment determined by?**

A The size of the application data
B The Maximum Segment Size (MSS)
C The size of the TCP header
D The size of the UDP header

11 **Which of the following best describes TCP?**

A Connectionless
B Connection-oriented
C Stateless
D Unreliable

12 **What does the term "port number" refer to in the transport layer?**

A A physical connection point
B A way to identify a device on a network
C A way to identify a specific application process
D A method for data encryption

13 **How does TCP ensure data integrity?**

A By using checksums
B By retransmitting lost segments
C By maintaining sequence numbers
D All of the above

14 **What is the role of the UDP header?**

A To ensure reliable delivery
B To provide basic information for data transmission
C To manage flow control
D To establish connections

15 **What is the function of a sequence number in TCP?**

A To keep track of the order of segments
B To identify the source application
C To manage flow control
D To ensure data encryption

16 **Which of the following is NOT a characteristic of UDP?**

A Connectionless
B Unreliable
C Flow control
D Lightweight

17 **What mechanism does TCP use to control congestion?**

A Acknowledgments
B Checksums
C Sliding window protocol
D Port numbers

18 **Which of the following statements about TCP's sliding window is true?**

A It helps in establishing connections
B It is used for error detection
C It allows multiple segments to be sent before requiring an acknowledgment
D It is a method for data encryption

19 **Which of the following is a feature of SCTP (Stream Control Transmission Protocol)?**

A Connectionless communication
B Multihoming support
C No congestion control
D Simple header structure

20 **What happens if a TCP segment is lost during transmission?**

A It is discarded
B The connection is terminated
C It is retransmitted
D The receiver sends a new segment

21 **In TCP, what does the "urgent" flag indicate?**

A The segment is a duplicate
B There is urgent data that should be prioritized
C The connection is closing
D The sender is congested

22 **How does TCP handle out-of-order segments?**

A It buffers them until they can be reassembled
B It discards them immediately
C It sends an acknowledgment for each segment
D It establishes a new connection

23 **Which transport layer protocol is commonly used for DNS queries?**

A UDP B TCP C SCTP D ICMP

Q **What is the primary function of the "ACK" (acknowledgment) in TCP?**

A To confirm receipt of data B To request additional data
C To terminate a connection D To signal congestion

Q **Which layer of the OSI model does the transport layer correspond to?**

A Network Layer B Application Layer
C Transport Layer D Data Link Layer

24 **What is the default maximum segment size (MSS) for TCP?**

A 1280 bytes B 1500 bytes
C 536 bytes D Varies based on the network

25 **What does the term "bandwidth delay product" refer to?**

A The maximum speed of a connection
B The amount of data that can be in transit in the network
C The time it takes for a packet to reach its destination
D The efficiency of data transmission

26 **Which of the following is a common application layer protocol that uses TCP?**

A DHCP B HTTP C ARP D ICMP

27 **In TCP, what is the purpose of the "FIN" flag?**

A To acknowledge data B To terminate a connection
C To request more data D To indicate congestion

28 **What is the typical window size in TCP?**

A 32 bytes
B 256 bytes
C 64 KB
D Varies depending on the network conditions

29 **Which mechanism is used by TCP to detect lost segments?**

A Sequence numbers B Retransmission timeout
C Checksums D Sliding window

30 **Which of the following statements is true about UDP?**

A It guarantees data delivery
B It establishes a connection before data transfer
C It is faster than TCP
D It has built-in error recovery

31 **What is the role of the "window" in TCP?**

A To control the flow of data
B To maintain connection state
C To verify segment integrity
D To manage application processes

32 **What happens during the "close" phase of a TCP connection?**

A A new connection is established
B Both sides acknowledge the connection termination
C Data is re-sent
D Flow control is stopped

33 **What is the purpose of the SYN flag in TCP?**

A To terminate a connection
B To initiate a connection
C To indicate urgent data
D To acknowledge receipt

34 **Which protocol is used for real-time applications, like video streaming?**

A TCP B FTP C UDP D SCTP

35 **What is the significance of the "timeout" period in TCP?**

A It determines how long a segment can remain in the buffer
B It helps decide when to retransmit lost segments
C It regulates flow control
D It signals the end of transmission

36 **What does the term "retransmission" refer to in TCP?**

A Sending a segment again after detecting loss
B Closing an existing connection
C Acknowledging receipt of data
D Establishing a new connection

37 **Which of the following best describes SCTP?**

A A simplified version of TCP
B A replacement for UDP
C A protocol that supports multi-streaming
D A connectionless protocol

38 **In TCP, what is the result of receiving three duplicate ACKs?**

A The connection is terminated
B A new connection is established
C Fast retransmit is triggered
D Flow control is initiated

Part 4 – Switching & Application Layer

28 multiple-choice questions · Circuit vs. packet switching, application-layer services and protocols

Switching Concepts

Q **A local telephone network is an example of a _____ network.**
A Packet switched
B Circuit switched
C Bit switched
D Line switched

Q **Most packet switches use this principle _____**
A Stop and wait
B Store and forward
C Store and wait
D Stop and forward

Q **The required resources for communication between end systems are reserved for the duration of the session between end systems in _____ method.**
A Packet switching
B Circuit switching
C Line switching
D Frequency switching

Q **In _____ systems, resources are allocated on demand.**
A packet switching
B circuit switching
C line switching
D frequency switching

Application Layer

Q **Which of the following is not an application layer service?**
A Network virtual terminal
B File transfer, access, and management
C Mail service
D Error control

2 **The packet of information at the application layer is called _____**
A Packet B Message C Segment D Frame

Q **Which one of the following is an architecture paradigms?**
A Peer to peer
B Client-server
C HTTP
D Both Peer-to-Peer & Client-Server

Q **Application developer has permission to decide the following on transport layer side**
A Transport layer protocol
B Maximum buffer size
C Both Transport layer protocol and Maximum buffer size
D None of the mentioned

Q **Application layer offers _____ service.**
A End to end
B Process to process
C Both End to end and Process to process
D None of the mentioned

Q **E-mail is _____**
A Loss-tolerant application
B Bandwidth-sensitive application
C Elastic application

Q **Pick the odd one out.**
A File transfer
B File download
C E-mail
D Interactive games

Q **Which of the following is an application layer service?**
A Network virtual terminal
B File transfer, access, and management
C Mail service
D All of the mentioned

Q **To deliver a message to the correct application program running on a host, the _____ address must be consulted.**
A IP
B MAC
C Port
D None of the mentioned

Q **Which is a time-sensitive service?**
A File transfer
B File download
C E-mail
D Internet telephony

Q **Transport services available to applications in one or another form _____**
A Reliable data transfer
B Timing
C Security
D All of the mentioned

Q **Electronic mail uses which Application layer protocol?**
A SMTP
B HTTP
C FTP
D SIP

Q **The _____ translates internet domain and host names to IP address.**
A domain name system
B routing information protocol
C network time protocol
D internet relay chat

Q **Which one of the following allows a user at one site to establish a connection to another site and then pass keystrokes from local host to remote host?**
A HTTP
B FTP
C Telnet
D TCP

Q **Application layer protocol defines _____**
A types of messages exchanged
B message format, syntax and semantics
C rules for when and how processes send and respond to messages
D all of the mentioned

Q **Which one of the following protocols delivers/stores mail to receiver server?**
A simple mail transfer protocol
B post office protocol
C internet mail access protocol
D hypertext transfer protocol

Q **Which one of the following is an internet standard protocol for managing devices on IP network?**

A dynamic host configuration protocol
B simple network management protocol
C internet message access protocol
D media gateway protocol

Q **Which one of the following is not an application layer protocol?**

A media gateway protocol
B dynamic host configuration protocol
C resource reservation protocol
D session initiation protocol

Q **Which protocol is a signaling communication protocol used for controlling multimedia communication sessions?**

A session initiation protocol
B session modelling protocol
C session maintenance protocol
D resource reservation protocol

Q **Which one of the following is not correct?**

A Application layer protocols are used by both source and destination devices during a communication session
B HTTP is a session layer protocol
C TCP is an application layer protocol
D All of the mentioned

Q **When displaying a web page, the application layer uses the _____**

A HTTP protocol
B FTP protocol
C SMTP protocol
D TCP protocol

Q **The IETF standards documents are called _____**

A RFC
B RCF
C ID
D DFC

Q **In the layer hierarchy as the data packet moves from the upper to the lower layers, headers are _____**

A Added
B Removed
C Rearranged
D Modified

Q **The structure or format of data is called _____**

A Syntax
B Semantics
C Struct
D Formatting

Part 5 – Reliable Data Transfer (rdt)

Match each description to its protocol version, then review the key statements

Protocol Matching

Q	**underlying channel perfectly reliable no bit errors no loss of packets**		
	rdt1.0	rdt2.0	rdt3.0
Q	**underlying channel may flip bits in packet checksum (e.g., Internet checksum) to detect bit errors**		
	rdt1.0	rdt2.0	rdt3.0
Q	**sender waits “reasonable” amount of time for ACK retransmits if no ACK received in this time**		
	rdt1.0	rdt2.0	rdt3.0

Key Statements

- **Acknowledgements (ACKs):** receiver explicitly tells sender that pkt received OK **(True)**
- **Negative acknowledgements (NAKs):** receiver explicitly tells sender that pkt had errors **(True)**
- Sender retransmits pkt on receipt of NAK **(True)**
- Stop and wait is sender sends one packet, then waits for receiver response **(True)**
- utilization is fraction of time sender busy sending **(True)**
- pipelining is sender allows multiple, “in-flight”, yet-to-be-acknowledged packets **(True)**

Part 6 – Essay

Worked delay, throughput and utilization problems with solutions

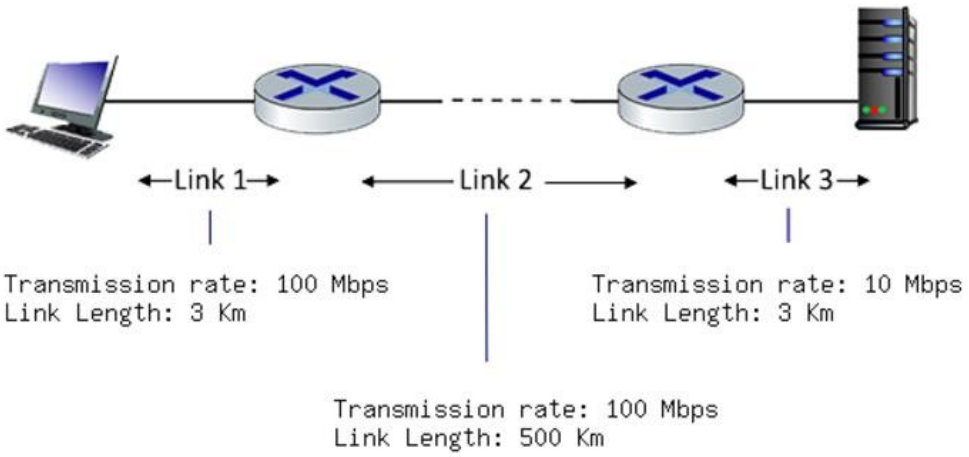
Nodal Delay

$$d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$$

- **Processing Delay:** Time taken to examine the packet's header and determine the packet's direction.
- **Queue Delay:** Occurs when packets have to wait in a queue before being transmitted.
- **Transmission Delay:** Time required to push all packet bits onto the link, calculated as packet length divided by transmission rate.
- **Propagation Delay:** Time taken for a bit to propagate from one router to the next, determined by the distance between routers and the propagation speed of the link.

Link Delay Problems

Assume the length of a packet is 4000 bits. The speed of light propagation delay on each link is 3x10^8 m/sec



Packet length = 4000 bits **Speed of light (c) = 3 x 10^8 m/s**

1. **What is the transmission delay of link 1?**

Transmission rate = 100 Mbps = 100 x 10^6 bps

Transmission delay = Packet length / Transmission rate $\frac{L}{R}$

Transmission delay = 4000 bits / (100 x 10^6 bps) = 4 x 10^{-5} seconds = 0.00004 seconds

2. **What is the propagation delay of link 1?**

Link length = 3 km = 3x10^3=3000 m

Propagation delay = Link length / Speed of light $\frac{D}{S}$

Propagation delay = 3000 m / (3 x 10^8 m/s) =1 x10^{-5} = 0.00001 seconds

3. **What is the total delay of link 1?**

Total delay = Transmission delay + Propagation delay

Total delay = 0.00004 seconds + 0.00001 seconds = 0.00005 seconds

4. What is the transmission delay of link 2?

Transmission rate = 100 Mbps = 100 x 10^6 bps
Transmission delay = 4000 bits / (100 x 10^6 bps) = $4x10^{-5}$ seconds

5. What is the propagation delay of link 2?

Link length = 500 km = 500000 m
Propagation delay = 500000 m / (3 x 10^8 m/s) = 1.66 $x10^{-3}$ seconds

6. What is the total delay of link 2?

Total delay = $4x10^{-5}$ + 1.66 $x10^{-3}$ = 1.7 x 10^{-3} = 0.0017 seconds

7. What is the transmission delay of link 3?

Transmission rate = 10 Mbps = 10 x 10^6 bps
Transmission delay = 4000 bits / (10 x 10^6 bps) = 4 $x10^{-4}$ seconds

8. What is the propagation delay of link 3?

Link length = 3 km = 3000 m
Propagation delay = 3000 m / (3 x 10^8 m/s) = 1 x 10^{-5} seconds

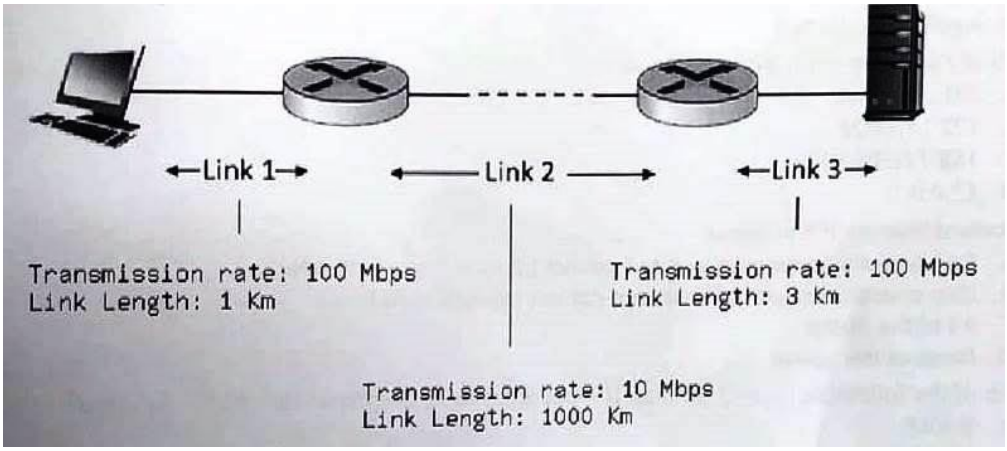
9. What is the total delay of link 3?

Total delay = 4 $x10^{-4}$ + 1 x 10^{-5} = 4.1 x 10^{-4} = 0.00041 seconds

10. What is the total delay?

Total delay = Total delay of Link 1 + Total delay of Link 2 + Total delay of Link 3
Total delay = 0.00005 + 0.0017 + 0.00041 = 2.16$x10^{-3}$ = 0.00216 seconds

1. Consider the figure below, with three links, each with the specified transmission rate and link length. Assume the length of a packet is 4000 bits. The speed of light propagation delay on each link is $3x10^8$ m/sec. What is the total delay from client to server? Round your answer to two decimals



L=4000 bit S=$3x10^8$ m/sec

Link 1	Link 2	Link 3
R=100 Mbps d= 1 km R=100 x10^6 d= 1 x10^3	R=10 Mbps d= 1000 km R=10 x10^6 d= 1000 x10^3	R=100 Mbps d= 3 km R=100 x10^6 d= 3 x10^3
T=L/R = 4000/(100 x10^6) = 0.00004 Prop =d/s = (1 x10^3)/(3x10^8) = 0.0000033333	T=L/R =4000/(10 x10^6) = 0.0004 Prop=d/s = (1000 x10^3)/(3x10^8) = 0.00333333333	T=L/R = 4000/(100 x10^6) = 0.00004 Prop=d/s = (3 x10^3)/(3x10^8) = 0.00001
Total Delay = 0.00004+0.0000033333+0.0004+0.00333333333+0.00004+0.00001= 0.003826		

Queuing & Nodal Delay

A packet is sent from router A to router B over a link with a distance 1500 km (propagation speed 4.5x10^8 m/s). When the packet arrives at router A, one other packet is halfway done being transmitted on this outbound link and three other packets are waiting to be transmitted. Suppose all packets are 2,500 bytes and the link rate is 3 Mbps.

> Transmission delay at the link = ? ms
> Propagation delay at the link = ? ms
> Queuing delay = ? ms
> The nodal delay from the time the packet enters router A until the packet arrives at router B = ? ms

Given Data

- Distance between routers A and B: $d = 1500 \text{ km} = 1,500,000 \text{ m}$
- Propagation speed: $s = 4.5 \times 10^8 \text{ m/s}$
- Packet size: $L = 2500 \text{ bytes} = 2500 \times 8 = 20,000 \text{ bits}$
- Link rate (bandwidth): $R = 3 \text{ Mbps} = 3 \times 10^6 \text{ bits/s}$

Step 1: Calculate Transmission Delay

The **transmission delay** is the time needed to push all the bits of a packet onto the link.

$$\text{Transmission Delay} = \frac{\text{Packet Size}}{\text{Link Rate}} = \frac{20,000 \text{ bits}}{3 \times 10^6 \text{ bits/s}}$$

$$\text{Transmission Delay} = \frac{20,000}{3,000,000} = 0.0067 \text{ seconds} = 6.67 \text{ ms}$$

Step 2: Calculate Propagation Delay

The **propagation delay** is the time it takes for the signal to travel from router A to router B.

$$\text{Propagation Delay} = \frac{\text{Distance}}{\text{Propagation Speed}} = \frac{1,500,000 \text{ m}}{4.5 \times 10^8 \text{ m/s}}$$

$$\text{Propagation Delay} = 0.0033 \text{ seconds} = 3.33 \text{ ms}$$

Step 3: Calculate Queueing Delay

When the packet arrives at router A:

- One packet is halfway done being transmitted, so it requires an additional $\frac{6.67}{2} = 3.335$ ms to finish transmission.
- Three additional packets are waiting in the queue, each requiring the full transmission delay of 6.67 ms.

The total **queueing delay** is:

$$\text{Queueing Delay} = 3.335 \text{ ms} + 3 \times 6.67 \text{ ms} = 3.335 + 20.01 = 23.345 \text{ ms}$$

Step 4: Calculate Nodal Delay

The **total nodal delay** is the sum of the transmission delay, propagation delay, and queueing delay.

$$\text{Nodal Delay} = \text{Transmission Delay} + \text{Propagation Delay} + \text{Queueing Delay}$$

$$\text{Nodal Delay} = 6.67 \text{ ms} + 3.33 \text{ ms} + 23.345 \text{ ms} = 33.345 \text{ ms}$$

Propagation vs. Transmission Delay

How long does it take a packet of length 1,000 bytes to propagate over a link of distance 2,500 km, propagation speed 2.5 x 10^8 m/s, and transmission rate 2 Mbps?

How long does it take a packet of length L to propagate over a link of distance d and transmission rate R bps?

Does this delay depend on packet length? Does this delay depend on transmission rate?

> Answer: Propagation delay= d/s = (2500 x 10^3) / (2.5 x 10^8) =0.01 sec 0.01x10^3= 10 msec
> L=1000 x 8=8000 bit R=2 x10^6=2000000 bit
> Transmission delay= L/R =8000/2000000 =0.004 sec= 0.004x10^3= 4 msec
> Total Delay time = 4ms + 10 ms = 14 ms

- **Does the delay depend on the packet length?**
  - Propagation delay: Does not depend on the packet length L; it depends only on d and s.
  - Transmission delay: Depends on the packet length L; longer packets take more time to be transmitted.
- **Does the delay depend on the transmission rate?**
  - Propagation delay: Does not depend on the transmission rate R; it is a function of d and s.

Transmission delay: Depends on R; a higher transmission rate reduces the transmission delay.

Throughput Problems

Suppose Host A wants to send a large file to Host B. The path from Host A to Host B has three links, of rates R1 = 500 kbps, R2 = 2 Mbps, and R3 = 1 Mbps.

a. Assuming no other traffic in the network, what is the throughput for the file transfer?

b. Suppose the file is 4 million bytes. Dividing the file size by the throughput, roughly how long will it take to transfer the file to Host B?

c. Repeat (a) and (b), but now with R2 reduced to 100 kbps.

> a) Consider given data: R1 = 500 kbps, R2 = 2 Mbps, and R3 = 1 Mbps
> The throughput for the file transfer=min {R1, R2, R3}
> =min {500 kbps, 2 Mbps, 1 Mbps}

=500 kbps
So, the throughput for the file transfer=500 kbps
b) Consider given data:
The file size= 4 million bytes
Convert million bytes to bits= 4000000*8 =32000000 bits.
From (a), Throughput for the file transfer=500 Kbps= $500*10^3$ =500000 bps

Time to transfer: $\frac{\text{File size (bits)}}{\text{Throughput (bps)}}$ =32000000 bits/500000 bps =64 seconds

c) Consider the given data:
Repeat (a) and (b), but now with R2 reduced to 100 kbps.
That means R2=100 kbps, R1 = 500 kbps, and R3 = 1 Mbps
The throughput for the file transfer=min {R1, R2, R3}
=min {500 kbps, 100 kbps, 1 Mbps}
=100 kbps
So, the throughput for the file transfer=100 kbps

Time to transfer: $\frac{\text{File size (bits)}}{\text{Throughput (bps)}}$ = 32000000 bits/100000 bps =320 seconds

Suppose two hosts A and B are separated by 10000 kilometers and are connected by a direct link of R=1Mbps. Suppose the propagation speed is 250000000 meters/sec. How long does it take to send a file of 400000 bits from A to B assuming that the file is divided into 10 packets with 40000 bits each? Assume that A does not send a packet until the preceding one has been received by B.

Sloution

-Two hosts (A and B) are connected using direct link.
-The distance between these two hosts A and B d= 10,000 km = $10{,}000x10^3$ m
-Transmission rate of the direct link between A and B R = 1 Mbps = 1×10^6 bps
-Propagation speed of the link between A and B S = 250000000 meters/sec =$25x10^7$
-The length of the file (L) = 400,000 bits
-The file is divided into 10 packets. So, the length of each packet =40,000 bits
-Calculate the transmission delay =L/R
= ($40{,}000 / 1 \times 10^6$) = 0.04 Second
-Calculating the propagation delay = distance/speed= d/s
= ($10{,}000x10^3 / 25x10^7$) = 0.04 Second
-Calculate the total time required to transmit 10 packets:
= n ($2d_{prop} + d_{trans}$) =10 (2x0.04+0.04) =10 (0.08+0.04)
=10 (0.12) =1.2 second

Throughput: Throughput refers to the amount of data per second that can be transferred between end systems.

$$\text{Throughput} = \frac{\text{Total Data Successfully Transferred}}{\text{Total Time Taken for Transfer}}$$

The throughput is generally the transmission rate of bottleneck link.

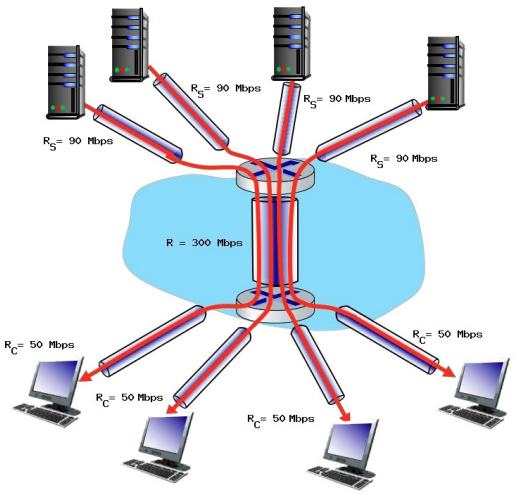
Shared-Link Bottleneck Scenarios

Consider the scenario shown below, with four different servers connected to four different clients over four three-hop paths.

The four pairs share a common middle hop with a transmission capacity of R = 300 Mbps.

The four links from the servers to the shared link have a transmission capacity of R_S = 90 Mbps.

Each of the four links from the shared middle link to a client has a transmission capacity of R_C = 50 Mbps.



1. What is the maximum achievable end-end throughput (in Mbps) for each of four client-to-server pairs, assuming that the middle link is fairly shared (divides its transmission rate equally)?

The maximum achievable end-end throughput is the capacity of the link with the minimum capacity, which is 50 Mbps

2. Which link is the bottleneck link? Format as Rc, Rs, or R

The bottleneck link is the link with the smallest capacity between R_S, R_C, and R/4.
The bottleneck link is Rc.

3. Assuming that the servers are sending at the maximum rate possible, what are the link utilizations for the server links (R_S)? Answer as a decimal

The server's utilization = $R_{bottleneck}$ / R_S = 50 / 90 = 0.55

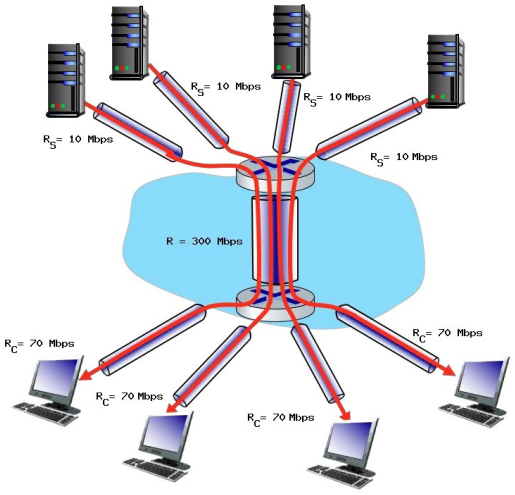
4. Assuming that the servers are sending at the maximum rate possible, what are the link utilizations for the client links (R_C)? Answer as a decimal

The client's utilization = $R_{bottleneck}$ / R_C = 50 / 50 = 1

5. Assuming that the servers are sending at the maximum rate possible, what is the link utilizations for the shared link (R)? Answer as a decimal

The shared link's utilization = $R_{bottleneck}$ / (R / 4) = 50 / (300 / 4) = 0.66

Consider the scenario shown below, with four different servers connected to four different clients over four three-hop paths. The four pairs share a common middle hop with a transmission capacity of R = 300 Mbps. The four links from the servers to the shared link have a transmission capacity of R_S = 10 Mbps. Each of the four links from the shared middle link to a client has a transmission capacity of R_C = 70 Mbps.



1. What is the maximum achievable end-end throughput (in Mbps) for each of four client-to-server pairs, assuming that the middle link is fairly shared (divides its transmission rate equally)?

The maximum achievable end-end throughput is the capacity of the link with the minimum capacity, which is 10 Mbps

2. Which link is the bottleneck link? Format as Rc, Rs, or R

The bottleneck link is the link with the smallest capacity between R_S, R_C, and R/4.
The bottleneck link is Rs.

3. Assuming that the servers are sending at the maximum rate possible, what are the link utilizations for the server links (R_S)? Answer as a decimal

The server's utilization = $R_{bottleneck}$ / R_S = 10 / 10 = 1

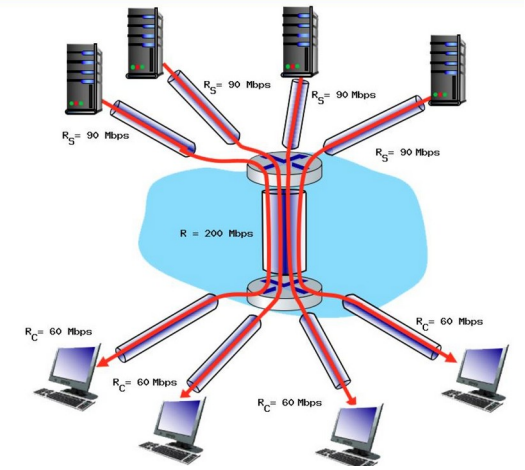
4. Assuming that the servers are sending at the maximum rate possible, what are the link utilizations for the client links (R_C)? Answer as a decimal

The client's utilization = $R_{bottleneck}$ / R_C = 10 / 70 = 0.14

5. Assuming that the servers are sending at the maximum rate possible, what is the link utilizations for the shared link (R)? Answer as a decimal

The shared link's utilization = $R_{bottleneck}$ / (R / 4) = 10 / (300 / 4) = 0.13

Consider the scenario shown below, with four different servers connected to four different clients over four three-hop paths. The four pairs share a common middle hop with a transmission capacity of R = 200 Mbps. The four links from the servers to the shared link have a transmission capacity of R_S = 90 Mbps. Each of the four links from the shared middle link to a client has a transmission capacity of R_C = 60 Mbps.



1. What is the maximum achievable end-end throughput (in Mbps) for each of four client-to-server pairs, assuming that the middle link is fairly shared (divides its transmission rate equally)?

The maximum achievable end-end throughput is the capacity of the link with the minimum capacity, which is 50 Mbps

2. Which link is the bottleneck link? Format as Rc, Rs, or R

The bottleneck link is the link with the smallest capacity between R_S, R_C, and R/4.
The bottleneck link is R.

3. Assuming that the servers are sending at the maximum rate possible, what are the link utilizations for the server links (R_S)? Answer as a decimal

The server's utilization = $R_{bottleneck}$ / R_S = 50 / 90 = 0.56

4. Assuming that the servers are sending at the maximum rate possible, what are the link utilizations for the client links (R_C)? Answer as a decimal

The client's utilization = $R_{bottleneck}$ / R_C = 50 / 60 = 0.83

5. Assuming that the servers are sending at the maximum rate possible, what is the link utilizations for the shared link (R)? Answer as a decimal

The shared link's utilization = $R_{bottleneck}$ / (R / 4) = 50 / (200 / 4) = 1

Utilization & Pipelining Example

Example: 1 Gbps link, 15 ms prop. delay, 8000-bit packet

Time to transmit packet into channel:

$$D_{trans} = \frac{L}{R} = \frac{8000\ bits}{10^9\ bits/sec} = 8\ microsecs$$

$$U_{sender} = \frac{L/R}{RTT + L/R}$$

$$= \frac{.008}{30.008}$$

$$= 0.00027$$

3-packet pipelining increases

$$U_{sender} = \frac{3L/R}{RTT + L/R} = \frac{.0024}{30.008} = 0.00081$$